Natural History

Alveary

Natural History: Grade 8

GRADE(S): 8

About the Course

Learners explore considerations from the history of microbiology and infectious disease. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 8

About Science: Grade 8
In level 8, learners study science in its historical, political, and cultural context, and with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 8
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 8
In level 8, learners study science in its historical, political, and cultural context with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

Natural History: Grade 8
For approximately eighth-grade students based on a balance of complexity and reading level. However, learners may be freely combined based on needs and preferences.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
8	Natural History: Grade 8 1 time/week 30 min	Invincible Microbe

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 8				
Natural History: Grade 8	General Science: Grade 8 Nature Walks & Scouting: Grades 1-8	General Science: Grade 8	General Science: Grade 8 Nature Notebook: Grade 8	Labs: Grade 8

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 8
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 8
Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

☐ Learn a few of your local species or varieties in connection with special topics.

Natural History: Grade 8
☐ Any term: a science museum with an exhibit on microbes

Term Prep & Teacher Tips

Natural History: Grade 8
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• 8 Tbsp cornstarch
• 1-2 c milk (Term 1)
• 3-5 c white vinegar
• paper towels
• water
• 1 egg (Term 1)
• 3 antimicrobial agents used at home (e.g., soap, hydrogen peroxide, rubbing alcohol, essential oils, etc.)

Reminders

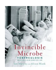
Natural History: Grade 8
☐ Note that the first lesson of Natural History is an activity, so be prepared with these materials and any others from the supply lists on day 1.
☐ Any type of milk is needed for Week 5. Make a note on your calendar if you will need to purchase this closer to the appropriate time.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ View Book List Details

Natural History: Grade 8


Invincible Microbe

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies
∞ View Supply List Details

Natural History: Grade 8


Prepared Microscope Slides

Household Items - Natural History: Grade 8


Slides and Coverslips


Student Microscope

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

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Natural History: Grade 8

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

Natural History: Grade 8

Click THIS text or scan the QR code for links.

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Term 1

WEEK 1

⬚ 30m Natural History: Grade 8 - Lesson 1

Cell Activity

▯ Materials: toothpick, glass slide and cover slip, candle, methylene blue stain, water, paper towel, microscope, egg, cup/glass, white vinegar

➔ INTRO
Most of your experience in natural history has been with Things that you can see around you. The invention of the microscope led to the discovery of a whole new world. This world of microscopic life is our focus this year. We begin by better understanding the microscopic cells from which you are made. First, you will make a slide of your cheek cells. Then you will set up an egg as a simple model of the cell.

➔ ACTIVITY
• Using a toothpick, gently rub the inside of your cheek.

• Now rub the toothpick on the surface of a glass slide, keeping the smear near the middle of the slide. Let this dry for a moment on a paper towel while you light your candle.

• Heat fix your slide by passing the slide through the flame 3-4 times. The underside of the slide should pass through the flame, but never stop. This will help your sample stick to the slide better. When you are finished, you can blow out the candle.

• Using protection, squeeze a drop of methylene blue stain onto your cheek smear. This stain will stain everything, so you MUST protect your clothing and anything else you do not want to be stained! Let the slide with the drop of stain sit for about 2 minutes.

• Gently rinse from the back of the slide with a little clean water and carefully blot dry. Do not rub your smear.

➔ OBSERVE & NARRATE
• Set a cover slip on your slide and examine it under the microscope, slowly increasing the power as you want to see more. Draw or diagram what you see in your science notebook.

➔ ACTIVITY
• Now, place an egg inside a cup or glass and cover with vinegar. Replace the vinegar every day for three days. Once the shell has been dissolved, the egg can be left on a paper towel until your next lesson.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ 30m Natural History: Grade 8 - Lesson 2

Cell Activity

▯ Materials: egg from last week, Prepared Microscope Slide Set, microscope, image link

PREP: Read Teacher Tip

➔ INTRO
Today, you will compare your egg model to the cheek cells that you observed under the microscope, compare these to a variety of animal cells, and then to some bacterial cells.

➔ OBSERVE, NARRATE, & DISCUSS
• Notice that your egg model has an outer membrane and a well-formed internal structure (the yolk) that is suspended in a matrix (the egg white). Compare to your cheek cells from last week. How are they similar? How are they different?

• This same basic structure is used to accomplish a great number of tasks. View slides 14-20 from your Prepared Microscope Slide Set. Try to identify these basic parts in each slide. Look carefully at slide 15. Red blood cells lose their nucleus so that they can carry oxygen more efficiently. You may also be able to find some white blood cells on this slide. How do they look different?

• View the Cellular Diversity Slides at the link to see a variety of bacterial cells. What do you notice about the bacterial cells in comparison? Can you still identify the same basic parts? How do the cells compare in size? Draw or diagram your observations in your science notebook
∞ Link: Cellular Diversity Slides

★ TEACHER TIP
If needed, be sure to pour your agar plates for next week according to the package instructions.
∞ **Video Link:** Pouring Agar Plates (2:51)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ 30m Natural History: Grade 8 - Lesson 3

Microbe Activity

▯ Materials: agar plates

➔ INTRO
Because bacteria are so small, humans were unaware of their existence or their place in our world for a very long time. One way to see them more easily is to grow a colony of them. You will use petri dishes with a kind of food, called agar, to do this. The dishes will be left open at various locations for several hours to see what kinds of microorganisms are in the environment at each location.

➔ ACTIVITY
• Choose three locations that you would like to test for the presence of bacteria. These might be

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

your classroom, the bathroom, the kitchen, the playground, etc.

• Take four of your agar plates and, keeping them closed, turn them upside down on your work surface. Label the bottom of each plate using a permanent marker. One plate will be labeled "control," while the remaining three will be labeled with your three locations.

• Leave the control plate closed and where it is while taking the three experimental plates to their locations. For each experimental plate, open the lid and leave the plate right-side-up in that location. You will leave it there for about three hours.

• After several hours exposed, collect each plate, replacing the lid. You may want to secure all of the plates with tape. Then leave them in a warm (but not hot) place for the next week.

WEEK 4

⬚ 30m Natural History: Grade 8 - Lesson 4

Microbe Activity

> Materials: agar plates from last week

PREP: Read Teacher Tip

→ INTRO
Today, you will examine your plates! Which location do you think will have the most microbial growth?

→ OBSERVE, NARRATE, & DISCUSS
• Retrieve your four plates and examine them. Draw or diagram them in your science notebook. Hopefully, your control plate is still clean. Which of your experimental plates has the most growth? Describe the appearance of the growth. Are there dots of separate colonies? Is the entire plate covered? Does the growth all look the same, or are there different types of microbes, indicated by different colors and textures?

• It is important to understand that even though we are surrounded by microorganisms all the time, we usually don't get sick. Many microorganisms are harmless or even beneficial. Look around your home or school, or talk to adults to find three ways that microorganisms are beneficial. Examples are cheese, yogurt, any fermented foods, and certain health supplements.

★ TEACHER TIP
Next week's activity requires a brief prep that requires any type of milk.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ 30m Natural History: Grade 8 - Lesson 5

Infectious Disease Activity

> Materials: video link, cups prepared with milk and cornstarch (see Teacher Tip), iodine drops

PREP: Read Teacher Tip

→ INTRO
Recall what you have learned so far. Some microorganisms can make us sick. Understanding these microbes helps us protect ourselves and care for others. Watch the linked video to see one experiment. Viruses are specifically mentioned in the video, but the same idea applies to many disease-causing germs. Then you will do a similar experiment and compare your results.
∞ Video Link: How Viruses (and other germs) Spread (8:46)

→ ACTIVITY
• There are 8-12 cups. Each cup represents the 'body fluid' of a person. We will simulate a typical daily interaction between two people by swirling two cups, pouring the fluid first into one and then the other cup, and finally dividing it again between the two cups. Each person should 'interact' with 3-4 others.

• Complete the interactions for the group.

• Only one was originally infected. How many do you think contracted our pretend illness? Can you see any difference by looking at the cups? Add 1-2 drops of iodine to each cup and swirl one last time.

→ OBSERVE, NARRATE, & DISCUSS
• Each person whose 'body fluid' changed color was infected. What percentage of the total number were infected? How does this compare to what you thought would happen? Can you determine who infected the rest of the party?

★ TEACHER TIP
Prep for this week's activity: a cup with about 2 Tbsp of milk should be prepared for each group member. If a student is working alone, then 8-12 cups could be prepared to simulate a group. Write names on the cups to differentiate. "Infect" one of the cups with 2 tsp of cornstarch and mix well.

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in the microscopic world? More interest/ease in discussing how scientists have developed that knowledge? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a microbiology exhibit.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ 30m Natural History: Grade 8 - Lesson 6

Meet TB

> Materials: Invincible Microbe

PREP: Read Teacher Tip

→ INTRO
Look at the title and cover. Have you ever heard of tuberculosis (TB)? TB is one of those disease-causing germs. This story is about its part in natural history.

→ READ, NARRATE, & DISCUSS

★ TEACHER TIP
There is an INTRO for this week's lesson, but plan to wean students from these prompts and train them to RECAP on their own. This will smooth their transition into high school coursework.

★ TEACHER NOTE
The age of the Earth and evolution are mentioned multiple times over p.1-6 and in the supplemental video. Teachers

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Term 1

Ch.1 p.1-7 "This is" - "they encountered."

• On p.4, the author says, "The germs began to evolve. What emerged was a different and even more dangerous microorganism..." Discuss how this happened in terms of "natural selection"?
∞ Video Link: Natural Selection (5:02)

should preread the lesson if they have concerns and consider how they would like to discuss or what other theories they might like to share.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ 30m Natural History: Grade 8 - Lesson 7
TB in the Ancient World

Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.1 p.8-13 "The existence" - "throughout Europe."

★ TEACHER NOTE
The age of the civilization is mentioned on p.8: "hundreds of thousands of years." Teachers might choose to allow it to pass by the way or to consider how they would like to discuss or offer alternative theories they would like to share.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ 30m Natural History: Grade 8 - Lesson 8
Medieval Medicine

Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.2 p.14-17 "For days" - "human urine."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

⬚ 30m Natural History: Grade 8 - Lesson 9
Renaissance Medicine

Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.2 p.17-22 "While superstition" - "to get worse."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• IMPORTANT DATES
Thermometer invented (1715)
Stethoscope invented (1818)

WEEK 10

⬚ 30m Natural History: Grade 8 - Lesson 10
Industrial Surge

Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.3 p.23-30 "It was a" - "mumbo jumbo."

• The author states that the Industrial Revolution caused tuberculosis to spread. Explain.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11

⬚ 30m Natural History: Grade 8 - Lesson 11
Catch Up

Materials: Invincible Microbe

PREP: Read Teacher Tip

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether microbiology or the relationship between infectious disease and society is more familiar or has possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 1

WEEK 12

⬚ **30m Natural History: Grade 8 - Lesson 12**
Exam Week

→ EXAMS
Answer question(s) related to course.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 2

WEEK 1

⬚ **30m Natural History: Grade 8 - Lesson 13**
The Sanatorium

Materials: Invincible Microbe

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.4 p.31-33 "Hermann Brehmer" - "when forced to."

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ **30m Natural History: Grade 8 - Lesson 14**
The Sanatorium

Materials: Invincible Microbe

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.4 p.35-39 "Some sanatoriums" - "hope of a cure."

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ **30m Natural History: Grade 8 - Lesson 15**
Patients are Persons

Materials: Invincible Microbe

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.5 p.40-44 "In 1937" - "lot of fun."

• Do you think the patients were people who usually behaved as described in the last paragraph of the reading? Explain their behavior.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **30m Natural History: Grade 8 - Lesson 16**
Patients are Persons

Materials: Invincible Microbe

➔ RECAP

➔ READ, NARRATE, & DISCUSS
Ch.5 p.44-50 "Of course" - "cure--forever."

• Describe a typical day in the life of a patient in a sanatorium. What do you think about this?

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ **30m Natural History: Grade 8 - Lesson 17**
The Cause

Materials: Invincible Microbe

PREP: Read Teacher Tip

➔ RECAP

➔ READ, NARRATE, & DISCUSS
Ch.6 p.51-55 "Robert Koch" - "economic stability."

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in the microscopic world? More interest/ease in discussing the relationship between infectious disease and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a microbiology exhibit.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• **IMPORTANT DATES**
Robert Koch identifies TB (1882)

WEEK 6

⬚ **30m Natural History: Grade 8 - Lesson 18**
Fighting the Germ

Materials: Invincible Microbe

➔ RECAP

➔ READ, NARRATE, & DISCUSS
Ch.6 p.55-60 "Mustering support" - "newfangled machine."

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• **IMPORTANT DATES**
X-Rays discovered (1895)
Hermann Biggs (1859-1923)

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Term 2

WEEK 7

⬚ **30m Natural History: Grade 8 - Lesson 19**
Fighting the Germ

▯ Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.6 p.60-64 "The influence" - "called outsiders."

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ **30m Natural History: Grade 8 - Lesson 20**
The Outsiders

▯ Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.7 p.65-70 "In November 1906" - "beds available."

• Explain how AMA practices and policy kept people of color out of medicine. What effect did this have on communities?

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

⬚ **30m Natural History: Grade 8 - Lesson 21**
The Outsiders

▯ Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.7 p.70-75 "From the late" - "cooks and drivers."

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• **IMPORTANT DATES**
The Great Migration (1915-1930)

WEEK 10

⬚ **30m Natural History: Grade 8 - Lesson 22**
The Outsiders

▯ Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.7 p.75-81 "Edythe Tate-Thompson" - "as outsiders."

• When you are reading, do you sympathize more easily with those who are sick with TB or those who are afraid of getting sick? Try to imagine you are part of whichever group you did NOT choose. Now explain the problems with which you are faced as part of that other group. How do you feel about those problems? What are some reasonable solutions to those problems?

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11

⬚ **30m Natural History: Grade 8 - Lesson 23**
Catch Up

▯ Materials: Invincible Microbe

PREP: Read Teacher Tip

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
Next term begins with an activity! If needed, plan to prepare agar plates according to the package directions.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether microbiology or the relationship between infectious disease and society is more familiar or has possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 12

⬚ **30m Natural History: Grade 8 - Lesson 24**
Exam Week

PREP: Read Teacher Tip

★ TEACHER TIP
Next term begins with an activity! If needed, plan to prepare agar plates according to the package directions.
∞ **Video Link:** Pouring Agar Plates (2:51)

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Term 2

→ **EXAMS**
Answer question(s) related to course.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 3

WEEK 1

⬚ **30m Natural History: Grade 8 - Lesson 25**
Antimicrobial Activity

> Materials: three antimicrobial agents used at home, three agar plates, permanent marker, sensitivity discs, small forceps/tweezers, gloves, cotton swabs, rubbing alcohol or hot soapy water

→ INTRO
Recall what is happening in the story of TB. Think of three cleaning agents or natural remedies you might use to get rid of germs. How well do they work? Let's find out in this activity.

→ ACTIVITY
• Turn all of the agar plates upside down and use a marker to divide them into quadrants. Write "Control" in one quadrant of each plate and label each of the remaining quadrants for each of the antimicrobial agents to be tested. So, plate 1 will have the control, and conditions A, B, and C in the 4 quadrants. Plates 2 and 3 will repeat the same sample set. When the plates are all labeled, turn them back over.

• With gloved hands, use a sterile cotton swab to rub the inside of your mouth for 10 seconds. Lift the lid of one plate just enough to evenly and gently rub the cotton swab over the entire surface. Then, recover the plate with the lid. Repeat for all three plates.

• Use rubbing alcohol to sterilize the forceps or wash them in hot, soapy water. Then pick up a single, sensitivity disc with the forceps and apply the first of your antimicrobial agents to the disc. Remove the lid of one plate and place the disc into the center of the appropriate quadrant. Press gently with the forceps, but do not damage the agar growth medium. Repeat for plates 2 and 3.

• Clean the forceps again and then repeat the last step for the 2 remaining antimicrobial agents and the control. The control will be a plain disc.

• When you are finished, turn all the plates back upside down and incubate in a warm (but not hot) location. You may wish to tape them closed. If you have time to observe them during the week, then do. If not, then they can wait until next week.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ **30m Natural History: Grade 8 - Lesson 26**
Antimicrobial Activity

> Materials: plates from last week

→ INTRO
Today, you will examine your plates! How well do you think your antimicrobial agents worked?

→ OBSERVE, NARRATE, & DISCUSS
• Retrieve your three plates and examine them. Draw or diagram them in your science notebook. Did your antimicrobial agents inhibit growth better than the control? Which of your agents worked the best? Does this mean that it works as well against all microbes? How could you find out?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ **30m Natural History: Grade 8 - Lesson 27**
The 'Cure'

> Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.8 p.82-86 "After announcing" - "drastic indeed."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **30m Natural History: Grade 8 - Lesson 28**
The 'Cure'

> Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.8 p.86-91 "In Spain" - "doctor-approved cures."

• Describe some of the ineffective "cures" that were sold. What are the problems with these "cures"?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ **30m Natural History: Grade 8 - Lesson 29**
The 'Cure'

> Materials: Invincible Microbe

PREP: Read Teacher Tip

→ RECAP

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in the microscopic world? More interest/ease in discussing the relationship between infectious disease and society? If not, consider observing and discussing together more often, exploring

Term 3

→ READ, NARRATE, & DISCUSS
Ch.8 p.91-96 "In 1912," - "in his arms."

extra helpings, or taking a field trip to a science museum with a microbiology exhibit.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ 30m Natural History: Grade 8 - Lesson 30

Antibiotics

Materials: Invincible Microbe

PREP: Read Teacher Tip

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.9 p.97-103 "In the autumn" - "wouldn't die."

★ TEACHER NOTE
The age of the Earth is mentioned on p.102: "millions of years." Teachers might choose to allow it to pass by the way, to decide how they would like to discuss, or to consider alternative theories they would like to share.

★ TEACHER TIP
Next week is an activity week!

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ 30m Natural History: Grade 8 - Lesson 31

Antibiotic Activity

Materials: three agar plates, permanent marker, antibiotic discs, small forceps/tweezers, gloves, cotton swabs, rubbing alcohol or hot soapy water

→ INTRO
You just read about the discovery of an antibiotic for use against TB. Let's repeat the same activity from a few weeks ago, but use antibiotic discs this time. How do you think they will compare to the other antimicrobial agents? Do you think all three antibiotics will work the same compared to each other?

→ ACTIVITY
• Turn all of the agar plates upside down and use a marker to divide them into quadrants. Write "Control" in one quadrant of each plate and label each of the remaining quadrants for each antibiotic to be tested. Plate 1 will have the control, and antibiotics A, B, and C in the four quadrants. Plates 2 and 3 will repeat the same sample set. When the plates are all labeled, turn them back over.

• With gloved hands, use a sterile cotton swab to rub the inside of your mouth for 10 seconds. Lift the lid of one plate just enough to evenly and gently rub the cotton swab over the entire surface. Then, recover the plate with the lid. Repeat for all three plates.

• Use rubbing alcohol to sterilize the forceps. Then pick up a single antibiotic disk with the forceps and apply the first of your antimicrobial agents. Remove the lid of one plate and place the disk into the center of the appropriate quadrant. Press gently with the forceps, but do not damage the agar growth medium. Repeat for plates 2 and 3.

• Clean the forceps with rubbing alcohol and then repeat the last step for the 2 remaining antimicrobial agents and the control. The control will be a plain disc.

• When you are finished, turn all the plates back upside down and incubate in a warm (but not hot) location. You may wish to tape them closed. If you have time to observe them during the week, then do. If not, then they can wait until next week.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ 30m Natural History: Grade 8 - Lesson 32

Antibiotics

Materials: plates from last week

→ INTRO
Today, you will examine your plates! How well do you think your antimicrobial agents worked?

→ OBSERVE, NARRATE, & DISCUSS
• Retrieve your 3 plates and examine them. Draw or diagram them in your science notebook. Did your antibiotics inhibit growth better than the control? How did they compare to each other? Why do we have different kinds of antibiotics? Can you compare these results with those obtained a few weeks ago on your antimicrobial agents? Why or why not?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 8

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QR

Term 3

WEEK 9

⬚ **30m Natural History: Grade 8 - Lesson 33**
Supergerms

Materials: Invincible Microbe

➔ RECAP

➔ READ, NARRATE, & DISCUSS
Ch.10 p.104-111 "In the records" - "again and again."

• Consider what you learned about natural selection back in Ch.1. Then explain the drug resistance and its solution noted on p.105-106: "The side effects" - "permanent cure."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10

⬚ **30m Natural History: Grade 8 - Lesson 34**
Supergerms

Materials: Invincible Microbe

➔ RECAP

➔ READ, NARRATE, & DISCUSS
Ch.11 p.112-117 "There was" - "twenty other people."

• Difficult situations, like the one in this book, can make it difficult for people to treat each other with compassion. Why is that? Could the people in this story have done better? If so, how?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11

⬚ **30m Natural History: Grade 8 - Lesson 35**
Supergerms

Materials: Invincible Microbe

PREP: Read Teacher Tip

➔ RECAP

➔ READ, NARRATE, & DISCUSS
Ch.11 p.117-122 "Surprisingly," - "more comfortably."

• Discuss the need for responsibility in science. Give examples from this or other books that you have read.

★ TEACHER NOTE
The age of the Earth is mentioned on p.102: "millions of years." Teachers might choose to allow it to pass by the way, to decide how they would like to discuss, or to consider alternative theories they would like to share.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether microbiology or the relationship between infectious disease and society is more familiar or has possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 12

⬚ **30m Natural History: Grade 8 - Lesson 36**
Exam Week

➔ EXAMS
Answer question(s) related to course.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 8

Examination

Term 1

- Invincible Microbe: Tell what you know about bacteria using any combination of words, drawings, or diagrams.

Term 2

- Invincible Microbe: Explain some of the challenges in combating TB in the early days of the epidemic.

Term 3

- Invincible Microbe: Describe how drug resistance occurs. OR Tell what scientists learned from the TB epidemic.